Deep Learning in Emergency Life-Saving Care: Continuous Prehospital Physiological Monitoring Data for Trauma Outcome Prediction

[redacted]

Abstract*— Prehospital trauma requires decisions to be made under time pressure with limited information. Continuous physiological monitoring is routinely recorded during transport. Still, it is rarely used for prediction because the data are irregularly sampled and heavily incomplete. This dissertation evaluates whether the monitoring record alone can support a patient's risk prediction. A fully connected encoder over individual intervals, a per-interval convolutional encoder, and a whole-window convolutional encoder were trained on data from 603 patients, and their latent representations were passed to downstream classifiers, against a raw-channel baseline. In-hospital mortality was predicted with a test AUROC of 0.857, massive transfusion with 0.730, and arrival lactate with 0.627. Compression provided no consistent benefit: the raw channels were strongest for mortality, whereas the most aggressively compressed representation was strongest for transfusion. An interesting finding is that measurement presence alone reached 0.718 on mortality and 0.662 on transfusion, indicating that a substantial part of any physiological signal reflects the clinician's decision to monitor it.

Keywords—clinical decision support, trauma, physiological time series, representation learning, autoencoder, LSTM

I. Introduction

Clinical decision support systems (CDSS) are computational tools designed to assist clinicians in making decisions under uncertainty, time pressure and incomplete information (Walter Costa et al., 2021). Early trauma care is an ideal setting for decision support of this kind. Following a traumatic injury, medics must assess the patient and choose treatment actions immediately upon arrival to the emergency room, without access to extensive laboratory or imaging investigations and, where the patient is unconscious, without a history (Pisirir, 2024). These decisions usually carry life-or-death consequences, yet early clinical assessment is known to be imperfect. Wohlgemut et al. (2024) found that experienced clinicians identified major haemorrhage with only moderate accuracy, misdiagnosing roughly one in ten patients. Accurate prediction before hospital arrival could allow receiving teams to prepare blood products, activate major haemorrhage protocols and mobilise theatre staff in advance of the patient's arrival.

One line of work has approached this by building decision support tools around information a clinician can supply directly (Marsden et al., 2025) (Fig.1). This paper evaluates whether continuous pre-hospital data collected in an emergency vehicle by a ZOLL X monitor can be used to predict similar outcomes.



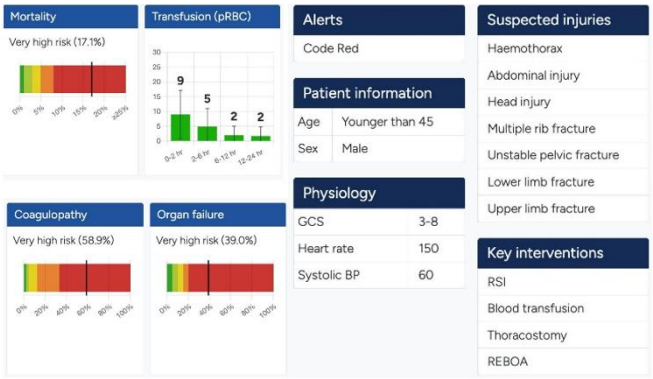


Fig. 1. Proposed AI-TRIPS interface (Queen Mary University of London, 2025)

The conditions under which this data is collected are not those of a ward or an intensive care unit. A trauma scene is chaotic: the patient must be rescued, moved onto a stretcher, loaded and unloaded, and treated en route in a moving vehicle. Sensors are attached when there is time and removed when they obstruct a procedure. As a result, channels are sampled at different rates, availability varies between patients and over the course of transport, and any given measurement may be absent for a big percentage of a recording. This makes the data difficult to process with conventional models.

This dissertation asks whether the continuous monitoring record can estimate patient outcomes and pursues four objectives in doing so. The first is to characterise the data itself: how completely each patient is monitored, how irregularly the channels are sampled, and whether the recorded pattern is informative on its own. The second is to develop a method to turn these irregular, heavily incomplete recordings into a compact numerical summary for each patient that a predictive model can use. This is done with an autoencoder, a network trained to compress a recording and then reconstruct it from the compressed representation, creating a summary. The third is to test whether these summaries predict two clinically consequential outcomes: in-hospital mortality and the need for massive transfusion. The last one is to establish how much of any predictive performance comes from the patient's physiology as opposed to the mere fact that a measurement was taken.

II. Related Work

A. Predicting trauma outcomes

A growing body of work demonstrates that physiological data available before hospital arrival can signal haemorrhage and indicate potential interventions. Several studies have shown that vital health records in combination with injury

characteristics can predict outcomes such as the need for blood transfusions (Pisirir, 2024; Sigle et al., 2026; Weidman et al., 2025). Sigle et al. (2026) is the closest analogue on the prediction task. They developed machine learning models that predict which patients would need a blood transfusion shortly after hospital arrival, based on routinely documented prehospital variables such as vital signs, injury patterns, and trauma mechanisms.

The line of research moves closer to the continuous-monitoring setting that motivates this project. Weidman et al. (2025) trained an ensemble model to predict the need for lifesaving interventions during prehospital care, using features derived from continuous physiological waveforms and vital sign patterns recorded from the moment the medical team reached the patient. The model predicted intervention within short windows during the journey and retained its performance when the signal used was recorded up to fifteen minutes beforehand. Guo et al. (2022) similarly used vital-sign time-series data, modelled with a gated recurrent unit (GRU), to produce dynamic predictions of massive haemorrhage over the subsequent one to three hours. Pisirir (2024) develops a model utilising Bayesian networks that predicts the volume of blood transfusion required based on the structured variables observed and recorded during transport or on arrival.

B. Missingness and irregular sampling in clinical time series

Working with electronic health records (EHR) is known to be obstructed by the irregularity of sampling, heterogeneity and sparsity of data (Xiang et al., 2026). Apart from that, the channels to be treated as a time series are sampled at different rates. For example, an electrocardiogram may contribute a thousand samples per minute while peripheral oxygen saturation contributes only tens.

One way to address this issue is rather than imputing gaps; an autoencoder can be supplied with a mask indicating which entries are genuinely observed and trained to reconstruct only those. This strategy is supported by the fact that missingness in clinical data is rarely random. Che et al. (2018) show that the occurrence of missing values typically correlates with the patient's state, a phenomenon they call informative missingness.

C. Unsupervised representation learning and autoencoders

The autoencoder provides a principled means of learning compact representations of high-dimensional unlabelled data. Originating as a non-linear approach to dimensionality reduction (Hinton and Salakhutdinov, 2006), an autoencoder is trained to reconstruct its input through a low-dimensional bottleneck, forcing the encoder to retain the most important structures.

In the clinical domain, autoencoders have been used to compress multimodal patient time series into representations useful for downstream tasks. Suresh, Szolovits and Ghassemi (2017) applied autoencoders to derive compressed representations of physiological features for patient phenotyping, demonstrating that learned representations can serve as an effective processing step. Bianchi et al. (2017) addressed the missing-data problem directly by combining an autoencoder with a kernel that accounts for missingness patterns, ensuring that the learned codes remain discriminative despite incomplete inputs. Xiang et al. (2026) used a dual-masked autoencoder which learns directly from incomplete time series by combining an intrinsic mask, which represents naturally missing values, with an augmented mask that hides a subset of observed values for reconstruction during training, processing only the unmasked tokens.

The choice of a sequence-to-sequence encoder-decoder is supported by evidence that such architectures yield representations that transfer effectively to downstream clinical prediction. Lyu et al. (2018) studied sequence-to-sequence models for clinical time series and found that the representations they learn are well-suited to subsequent prediction tasks. Unsupervised representations can reduce reliance on labelled data, which is difficult to obtain in clinical settings.

D. Sequence models for clinical prediction

Once a sequence of latent representations has been produced, a model capable of capturing temporal dependencies is required to map it to a prediction of patient state. The long short-term memory (LSTM) network by Hochreiter and Schmidhuber (1997) is a recurrent architecture suitable for this purpose. The sequence-to-sequence framework of Sutskever, Vinyals and Le (2014) generalised the use of such recurrent encoders and decoders to map between sequences, and underpins much of the representation-learning work discussed above. Following this is the proposed architecture, in which an LSTM takes the ordered sequence of interval-level latent representations produced by the encoder as input.

E. Convolutional architectures

Convolutional autoencoders have been applied to physiological time series, with Thill et al. (2021) showing that a temporal convolutional autoencoder can learn compressed representations of ECG recordings in an unsupervised setting. The pairing of a convolutional autoencoder with a downstream recurrent model was proposed by Zhao et al. (2019), who reported lower forecasting error for a CAE-plus-LSTM pipeline than for an LSTM alone on both traffic-flow and industrial fault data. Their proposed explanation is that reconstruction through the convolutional autoencoder reduces high-frequency noise, so that the LSTM operates on smoother input.

F. Research gap

Where these methods have been applied to patient data, it has almost entirely been in the intensive care unit, where recordings are noisy but also long, dense and produced with a consistent set of channels. A prehospital recording lasts minutes, is interrupted whenever treatment takes priority over monitoring, and contains only the channels the team had time to attach. This is the gap the research addresses.

The representations of such recordings are required to serve two purposes simultaneously. They need to be able to summarise the observed physiology of a recording window with fixed dimensions and do so in a way that is separable from the measurement presence pattern. A masked-reconstruction autoencoder was adopted because it satisfies both. Reconstruction error is evaluated only where the mask indicates a genuine observation, so unmeasured entries contribute no gradient, and no imputation is required. The bottleneck of the autoencoder fixes the output dimensionality, and because the objective is unsupervised, the encoder can be fitted on all recordings, even those missing outcome labels. A further property follows from keeping the mask separate from the values. It becomes possible to separate the contribution of the physiological signal from that of the decision to measure this signal.

III. Dataset

The data set consists of recordings for 606 patients. There are up to 10 files for each physiological channel associated with each patient; each file contains a time series for one measure, where, most of the time, each row represents a 30-second interval. The measures are listed in Table I. A separate table records clinical data for each patient, including arrival lactate, mortality, and blood transfusion volume. These labels are used for the prediction tasks in Section V.

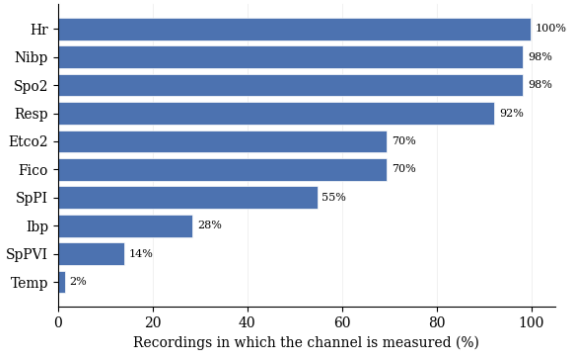


Fig. 2. Channel availability

TABLE I.
PHYSIOLOGICAL CHANNELS AVAILABLE

Channel	Description	Units	Notes
Heart rate (Hr)	Heart rate measured continuously	Beats per minute (bpm)	Most frequently present channel; used as the reference signal for defining each patient's analysis window.
Peripheral oxygen saturation (SpO_2)	Percentage of haemoglobin saturated with oxygen	%	Well populated but exhibits little variability, typically between 95–100%.
Non-invasive blood pressure (NIBP)	Oscillometric cuff blood pressure (systolic, diastolic, mean arterial pressure)	mmHg	Readings are forward-filled between cuff inflation cycles; treated as three separate channels.
End-tidal carbon dioxide ($EtCO_2$)	Partial pressure of CO_2 at the end of exhalation	mmHg	Requires capnography and an airway or sampling line.
Respiratory rate (Resp)	Breathing frequency	Breaths/min	Continuously monitored respiratory rate.
Invasive blood pressure (IBP)	Arterial catheter blood pressure (systolic, diastolic, mean arterial pressure)	mmHg	Continuous waveform available only for the most severely injured patients; treated as three separate channels.
Fractional inspired CO_2 ($FiCO_2$)	Concentration of inhaled carbon dioxide	kPa	Normally near zero; increases with rebreathing.
Perfusion index (SpPI)	Pulse oximetry-derived measure of peripheral pulse strength	%	Derived from pulse oximetry.
Pleth variability index (SpPVI)	Respiratory variation in the perfusion index	%	

A. Interior valid zeros

During the exploratory data analysis, an anomaly was noticed: a channel's status flag reported a reading as valid even though the value was zero. These were named interior valid zeros because they occur between the first and the last genuine reading of the recording. The phenomenon was most commonly observed in the $EtCO_2$ channel (34.9% of recordings) and the invasive blood-pressure channels (11-16%). The initial explanation for this was the measurement mechanism: $EtCO_2$ and invasive blood pressure are obtained through sampling lines and arterial catheters that are prone to being displaced while the patient is in an emergency vehicle. However, a deeper statistical analysis showed that the explanation might not be this straightforward.

To assess whether these zeros are simply a measurement artefact or carry clinical information, patients whose recordings were affected were compared with those whose were not. The comparison is restricted throughout to patients who have the channel in question, so that it reflects a difference between users of the same monitoring rather than the tendency for sicker patients to receive invasive monitoring in the first place.

TABLE II.
CLINICAL PROFILE OF PATIENTS WITH AND WITHOUT INTERIOR VALID ZEROS IN THE ETCO2 CHANNEL

Descriptor	Affected	Unaffected
Died	30% (42/138)	16% (41/252)
Rapid sequence intubation	50% (74/147)	34% (93/274)
Thoracostomy	37% (55/147)	26% (70/274)
Code Red activation	63% (93/147)	64% (176/274)
Head AIS	1.0 [0.0-4.0]	0.0 [0.0-3.0]
ISS	26.0 [16.5-49.0]	29.0 [20.0-42.5]
Scene-to-hospital (min)	50.0 [35.5-67.0]	44.0 [27.0-60.8]
Blood products 24 h (units)	4.0 [2.0-8.0]	3.0 [1.0-6.0]

[a.] Restricted to the 421 patients with capnography: 147 affected, 274 unaffected. Binary variables are shown as % (n/N); continuous variables as median [IQR].

The affected patients form a recognisably different clinical group: nearly twice as likely to die (30% versus 16%), substantially more likely to require rapid sequence intubation (50% versus 34%) and thoracostomy (37% versus 26%), longer at scene (median 50 versus 44 minutes), and receiving more blood products. Despite nearly double the mortality, affected patients had comparable injury severity scores (median ISS 26 versus 29). The interior zeros pattern therefore captures something other than the mere consequences of a more severe injury.

For the invasive blood-pressure channels, the pattern is weaker and less consistent. Among the 172 patients with invasive monitoring, the 42 affected showed higher rates of intubation (62% versus 46%), greater head injury severity (median Head AIS 2.0 versus 1.0) and longer scene-to-hospital times (62 versus 53 minutes), but mortality was identical between the groups (31% in both).

Two consequences follow. First, interior valid zeros are excluded from the reconstruction target, so the autoencoder is not trained to reproduce physiologically implausible zero readings. Second, because the pattern of these dropouts is associated with clinical severity rather than being random, the missingness pattern itself is retained as a potential source of signal, and its contribution is assessed separately by ablation for each model.

B. *Windowing and preprocessing*

Eleven of the available measures were carried into the analysis (Table III). The three non-invasive and three invasive blood-pressure measures are treated as separate channels, since each is recorded independently. Fractional inspired CO_2 was excluded from the analysis because in this dataset it is almost entirely zero and therefore carries little information. The pleth variability index was not included because it is derived from the perfusion index, and the two would provide largely redundant information; only the perfusion index was retained.

TABLE III.
CHANNELS INCLUDED IN THE ANALYSIS

Channel	%[b]
Hr	92.7
SpO_2	72.7
NIBP systolic	40
NIBP diastolic	39.9
NIBP mean	39.8
IBP systolic	19.2
IBP diastolic	19.5
IBP mean	19.5
$EtCO_2$	52.8
Respiratory rate	67.6
SpPI	40.4

[b] per channel % intervals observed

Each patient is represented by a 10-minute interval preceding the last valid heart rate reading, since heart rate is the most consistently available channel. This window is short relative to the recordings themselves, which have a median duration of 70 minutes (Fig. 3), so all but three of the 606 recordings are long enough.

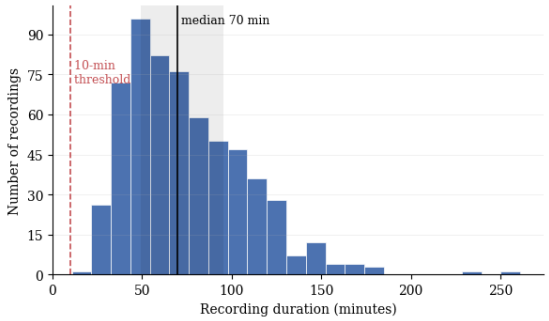


Fig. 3. Recordings duration distribution

A reading is treated as genuine only if the status field is "valid", its value is non-zero, it lies within a physiologically plausible range for that channel, and it is not an exact duplicate of an existing time–value pair. The window is divided into twenty 30-second intervals. Each interval is recorded as a vector $x \in \mathrm{R}^{11}$ a binary mask $m \in \{0,1\}^{11}$ recording which channels were observed. Each patient therefore yields two 20 x 11 arrays.

Patients are split 70/15/15 into training, validation and test sets (422, 90 and 91 patients). Each channel is divided by the 95th percentile of its observed training values, and the result multiplied by the mask, so missing entries remain exactly zero.

IV. ARCHITECTURE CHOICES

A. *Fully connected autoencoder*

The autoencoder learns a compact representation of the patient's physiological state within each time interval. Its role is to compress the multi-channel vector of each interval into a small latent vector so that the resulting sequence of latent vectors can be used as an input to a sequence model. The intervals are pooled across patients, so each of the 20 intervals of each patient is presented to the model as an independent training sample.

The autoencoder consists of an encoder and a decoder, each a feed-forward network with one hidden layer of 16 units and ReLU activation. The architecture design can be found in Appendix A (Fig.1). The encoder maps the interval vector to a latent representation $z \in \mathrm{R}^k$, and the decoder maps it back to a reconstruction $\hat{x} \in \mathrm{R}^C$, where k is the number of latent representations and C is the number of channels:

$$z = W_2\,\sigma(W_1 x + b_1) + b_2$$
$$\hat{x} = W_4\,\sigma(W_3 z + b_3) + b_4$$

where W_1.. W_4 and b_1.. b_4 are learned weights and biases, and the latent dimension $k < C$ enforces compression through the bottleneck.

The model is trained to minimise the reconstruction error over the observed channels only. Over a set of intervals B (a mini-batch during training, or the full pooled set when the objective is evaluated), the masked reconstruction loss is

$$\mathcal{L}(\mathcal{B}) = \frac{\sum_{n\in\mathcal{B}}\sum_{c=1}^{C} m_{n,c}\left(\hat{x}_{n,c} - x_{n,c}\right)^2}{\sum_{n\in\mathcal{B}}\sum_{c=1}^{C} m_{n,c} + \varepsilon}$$

with epsilon = 1e-8. The mask excludes missing channels from the numerator, and the denominator normalises by the total number of observed channel values. Because both sums accumulate across the whole set rather than within each interval, intervals in which many channels were recorded contribute proportionally more to the gradient than sparsely observed ones (the difference in observation rates is shown in Table III). The same objective is used for all three encoders, so differences in reconstruction error between them are attributable to the architectures alone.

B. *Per-interval convolutional autoencoder*

The dense autoencoder treats each 30-second interval as an independent observation. Any temporal structure in the data, for example, a heart rate that climbs across several consecutive intervals, must therefore be recovered downstream by the sequence model. The convolutional encoder addresses this by convolving along the time axis, so that each latent vector is a function of a short neighbourhood of intervals.

The model consumes the same arrays as the dense autoencoder, but whole patient windows are now presented to it rather than separate intervals. The encoder is a stack of two one-dimensional convolutions applied over the time axis, with the channel axis treated as the feature axis:

$$H = \sigma\left(\mathrm{Conv1D}_{C\to 16}(X^{\top})\right), \quad Z = \mathrm{Conv1D}_{16\to k}(H)$$

Both convolutions use a kernel of width 3 with padding 1, so the time axis is preserved at length T. Stacking two such layers gives each latent vector a receptive field of five intervals, that is 2.5 minutes of physiology centred on its own position. The decoder mirrors this with two transposed convolutions, $k \to 16 \to C$, again at kernel width 3 and padding 1. ReLU is applied only after the first layer of each network; the latent and output layers are linear. The architecture design can be found in Appendix A (Fig.2). Therefore, the encoder produces one latent vector per interval, preserving the sequence structure for the downstream model to use as input.

C. Whole-window convolutional autoencoder

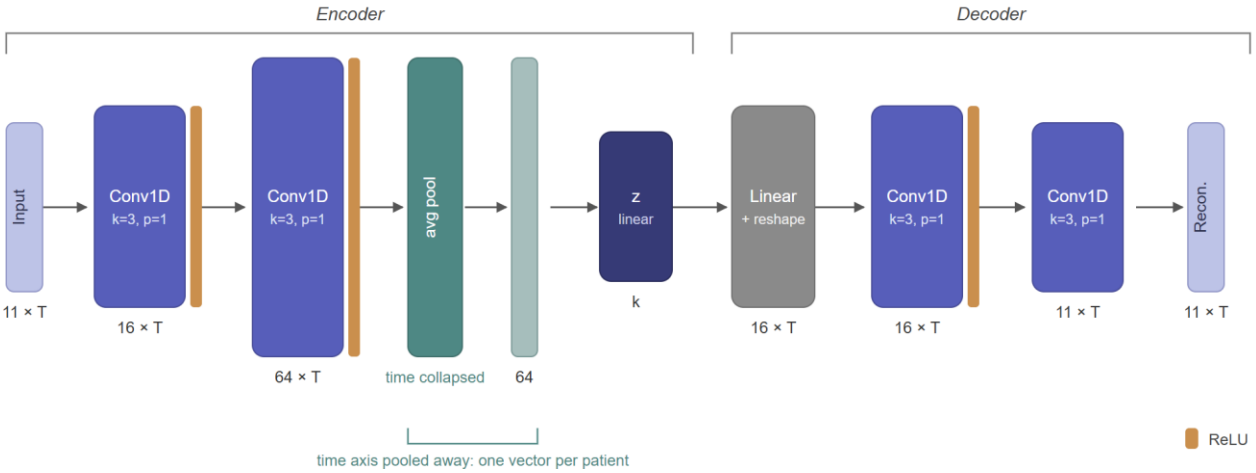


Fig. 4. Whole-window convolutional autoencoder architecture

A second convolutional encoder was implemented to test whether the per-interval representation is necessary, or whether a single vector summarising the entire window is enough. It differs from the encoder above in one respect: the time axis does not survive the bottleneck. The convolutional stack is widened to $C \to 16 \to 64$, again at kernel width 3 with padding 1, so the time axis is still preserved through the convolutions. An adaptive average pooling layer then collapses the twenty intervals to one, reducing the $(64 \times T)$ feature map to a single 64-dimensional vector, which a linear layer maps to the patient latent $\boldsymbol{z} \in \mathrm{R}^{k}$.

$$H = \mathrm{Conv1D}_{16\to 64}\left(\sigma\left(\mathrm{Conv1D}_{C\to 16}(X^{\top})\right)\right),$$
$$z = W\,\mathrm{pool}(H) + b$$

with ReLU applied after each convolution and the linear layer left inactivated. The decoder expands the latent to $16 \times T$ through a linear layer, which is reshaped to a feature map and passed through two further convolutions, $16 \to 16 \to C$, reconstructing the full $T \times C$ window (Fig. 4).

This is a substantially stronger bottleneck than the per-interval encoder, where the model reduces 11 channels to k at each of twenty time steps, retaining $20k$ values per patient, this one reduces the entire 20×11 window to k values, a compression of roughly 24-fold at $k = 9$.

D. Downstream classifiers

The fully connected and convolutional autoencoders that preserve the time axis produce a sequence of 20 vectors per patient, which is then passed to a single-layer LSTM classifier. The LSTM reads the intervals in order, maintaining a hidden state that is updated at each state, and the final hidden state is taken as a fixed-length summary of the whole window. A linear layer maps this vector to a single logit, and a sigmoid converts it to a predicted probability.

$$h_t = \mathrm{LSTM}(i_t, h_{t-1}, c_{t-1}), \quad t = 1, \ldots, T,$$
$$\hat{y} = \sigma_s(w^{\top} h_T + b)$$

$\mathrm{i_t}$ is the input at interval t (the latent vector, the mask, or their concatenation), $\mathrm{h_t} \in \mathrm{R^d}$ is the hidden state with $d \in \{16,32,64\}$, c_t the cell state, both initialised to zero. σ_s is the logistic sigmoid.

Because the whole-window encoder produces one vector per patient rather than a sequence, no recurrent classifier can be applied to it. The downstream model is instead a logistic regression on the patient latent, with an L2 penalty whose strength is treated as a tunable hyperparameter.

V. Training Procedure

A. Autoencoder training

All three autoencoders are trained on intervals (windows) from training-set patients only, with the masked reconstruction error monitored on held-out validation patients. All three are optimised with Adam at a learning rate of 1e-3 for 120 epochs.

The unit of a mini-batch differs depending on the input the encoder consumes: the dense autoencoder draws 256 intervals from the pooled set of training intervals, while both convolutional encoders draw 64 whole patient windows. Every encoder is trained from six random seeds at each latent dimension; the model from the median-performing seed is retained.

B. Latent dimensions

The dense and per-interval convolutional autoencoders were each swept over latent dimensions k = 2 to 10, and the whole-window encoder over k = 5 to 19. For the dense autoencoder and the per-interval convolutional encoder, masked validation error falls monotonically across the sweep. In both cases, the minimum lies at the largest value swept. Averaged over six seeds, the whole-window encoder's masked validation error varied non-monotonically, within the range of seed-to-seed variation. The sweep therefore provides no basis for selecting the latent dimension independently. Therefore, the latent dimensions were fixed at k = 9 for all three encoders, which made the architectures easier to compare.

C. Sequence classifier

Training minimises binary cross-entropy on the logits, optimised with Adam at a learning rate of 1e-3 for 200 epochs. Each epoch is a single gradient step computed over all labelled training patients. Patients with a missing label are excluded, leaving 559 patients for mortality (90 deaths) and 602 for massive transfusion (108 positives).

The LSTM is tuned over the number of hidden units {16,32,64}. For each candidate, the model is trained with six random seeds and evaluated by AUROC on the validation set. The width with the best mean validation AUROC is then retrained on the same six seeds and evaluated on the test patients. This procedure is repeated for each combination of task and input representation. AUROC is used instead of accuracy because both outcomes are heavily imbalanced.

D. Whole-window classifier

For the whole-window convolutional encoder, whose patient-level latent is classified by logistic regression, the L2 regularisation strength is chosen from the seven-point grid $C \in \{0.01, 0.03, 0.1, 0.3, 1, 3, 10\}$ by five-fold cross-validation on the training set. Each model is scored by AUROC, where the features are standardised inside the cross-validation pipeline. Because the encoder is retrained at each seed, the grid search is repeated within each of the six seeds.

E. Calibration

Each model is assessed for calibration, since it may rank patients well but produce risk estimates that are poorly scaled. For each encoder and each task, the physiology-only model is refitted at the validation-selected hidden size, using the same six seeds as for the AUROC. The Brier score is reported as a mean and standard deviation over those seeds, against a base-rate baseline.

VI. Results

Three outcome variables were evaluated. In-hospital mortality, blood transfusion (amount of blood products administered within 24 hours), and lactate were measured immediately upon the patient's arrival at the hospital.

Downstream classification was first approached with simpler models. Each patient's latent sequence was flattened into a single feature vector and passed to logistic regression and a random forest. Blood transfusion was defined as a 2-class variable, with patients divided into those who received 10 or more units of blood products. It was first tested as a 3-class problem (none, moderate, massive); however, simpler models have collapsed onto the majority class, with recall of only 5.6% for the no-transfusion group and 11.1% for the massive group. Lactate was also defined as a 2-class variable with 4 mmol/L as a threshold. It was first treated as a regression problem, but the downstream models did not perform better than predicting the training median. A linear model returned a test MAE of 5.57 against a median-prediction baseline of 3.26, and a random forest reached 3.20, with an R^2 of 0.043. It was therefore reformulated as a binary classification problem. However, reformulating lactate as a classification task did not recover the signal. Following the transfusion result, where the whole-window encoder was the strongest model, lactate was evaluated using that encoder alone. It reached a test AUROC of 0.627 ± 0.033 on the binary formulation.

A. Discrimination results

Performance is reported as the area under the receiver operating characteristic curve (Fig. 5). Mortality is evaluated on 84 test patients and massive transfusion on 91 after excluding the patients with a missing label. Comparisons are therefore made within a task rather than across them. The results are summarised in Table IV. On mortality, the raw-channel baseline is the strongest model at 0.857, and the three encoders rank in order of how much they compress the data. The dense encoder, which reduces eleven channels to nine, loses the least on discrimination, while the whole-window encoder, which reduces the entire window to nine values, loses the most. The gap between the raw baseline and the dense autoencoder is within roughly one standard deviation, so neither is clearly better than the other at discrimination.

The ordering reverses on massive transfusion. Here the whole-window encoder is the strongest model at 0.73, ahead of the raw baseline and both sequence encoders. Mortality appears to depend on details within the window that is destroyed by the averaging, whereas massive transfusion appears to benefit from summarisation, especially the patient-level one that the whole-window encoder is trained on.

TABLE IV.
TEST AUROC (MEAN +/- S.D. OVER SIX SEEDS) BY ENCODER AND INPUT REPRESENTATION

Task	Input	Dense AE	Conv AE	Conv[c] AE	Raw data
Mortality (n = 84)	Physiology only	0.810 +/- 0.073	0.771 +/- 0.032	0.736 ± 0.086	0.857 ± 0.026
	Mask only	0.718 +/- 0.024			
	Physiology + mask	0.778 +/- 0.046	0.735 +/- 0.030		0.784 ± 0.019
Transfusion (n = 91)	Physiology only	0.666 +/- 0.022	0.661 +/- 0.032	0.730 ± 0.039	0.674 ± 0.014
	Mask only	0.662 +/- 0.018			
	Physiology + mask	0.691 +/- 0.025	0.670+/- 0.032		0.656 ± 0.016
Lactate (n = 67)	Physiology only			0.627 +/- 0.033	

[c] Whole-window convolutional encoder

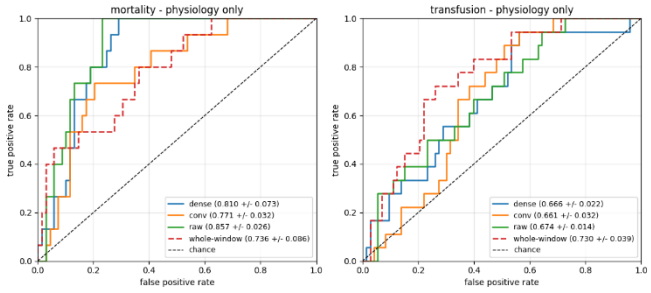


Fig. 5. AUROC comparison

The three input conditions separate the physiological signal from its presence. The central task of such a design was to determine whether the physiological values contributed anything beyond the monitoring pattern. The mask-only condition is directly evaluated by the sequence model, bypassing the encoder. Therefore, it is the same across all models, with AUROCs of 0.718 for mortality and 0.662 for

transfusion. It establishes a baseline that physiology-based models must exceed.

That baseline is substantial, confirming that the decision to attach a certain channel is in itself informative about how unwell the clinician judged the patient to be. The physiology nonetheless adds to it. All four physiology-based models exceed the mask-only baseline of 0.718, but only the raw channels and the dense encoder do so substantially; the convolutional (0.771) and whole-window (0.736) encoders sit close enough to the baseline. This means that at least two of the models are not merely recovering clinical judgement from the monitoring pattern. For massive transfusion, the margin is negligible for the sequence models, and only the whole-window encoder (0.730) clearly improves on the baseline (0.662). Combining the two representations does not aid the models: it improves on physiology alone in only two of six conditions and degrades mortality performance in every case.

B. Calibration results

Table V reports Brier scores against a base-rate baseline. On mortality, every model improves on the baseline of 0.147, the best reaching 0.119. On massive transfusion, the picture is less uniform: the whole-window encoder performs best at 0.146 against a baseline of 0.159, while the per-interval convolutional encoder is the only model in the study that fails to improve on its baseline at 0.161. Across models, the reliability curves share a common shape on mortality: predicted risk is close to zero in the lowest bins, where almost no deaths occur, and the risk is understated in the highest bins. On massive transfusion, the curves are flatter and noisier, with no consistent direction of miscalibration across models (Appendix C).

TABLE V.
BRIER SCORE (MEAN ± S.D. OVER SIX SEEDS)

Task	Base line	Dense AE	Conv AE	Conv[d] AE	Raw data
Mortality	0.147	0.123 +/- 0.013	0.134 +/- 0.01	0.131 +/- 0.01	0.119 +/- 0.005
Transfusion	0.159	0.153 +/- 0.003	**0.161 +/- 0.005**	0.146 +/- 0.003	0.152 +/- 0.002

[d.] Whole-window convolutional encoder

VII. DISCUSSION

A. Main findings

The expectation motivating this work was that compressing an irregular, heavily incomplete monitoring record into a compact latent summary would make it more usable by a downstream predictive model. The results show that what matters is not how much is compressed but which structure is discarded. The dense and per-interval convolutional encoders produce representations of identical size, differing only in that the latter smooths each latent vector over a five-interval receptive field, and that smoothing alone costs four points of AUROC on mortality. The whole-window encoder, which removes within-window variation altogether, loses most on mortality but is the strongest model in the study on massive transfusion. The raw normalised channels remain the strongest condition on mortality overall. Compression as a preprocessing step therefore does not necessarily add to the signal in this data, and whether it helps depends on the outcome and on the downstream classifier.

B. Why compression did not aid mortality

Three considerations can explain the mortality result. The first is that the input was not high-dimensional enough for the compression to pay off. Autoencoders earn their place where the input space is large relative to the information it carries, so that a bottleneck removes redundancy. Here each interval is described by eleven channels, of which a median of roughly five are observed, so reducing eleven dimensions to nine discards little redundancy.

The second is that the reconstruction objective is trained without reference to the outcome; minimising squared error recovers the central tendency while treating the fluctuations around it as noise. Short-term variability in vital signs is precisely the kind of structure this suppresses, and it is also the kind of structure associated with physiological instability.

The third follows from the form of a masked loss itself. Because the numerator and denominator both accumulate across the whole set rather than within each interval, every observed channel value contributes equally to the gradient. Reconstruction quality is therefore dominated by the channels that are most frequently present, heart rate at 92.7% of intervals and peripheral oxygen saturation at 72.7%, rather than by the channels that carry the most predictive signal. The encoder ends up allocating its capacity according to how often a channel is measured, not according to what the measurement means.

It should be said that the gap between the raw baseline and the dense autoencoder is within roughly one standard deviation. Therefore, the autoencoder is not clearly worse; it simply does not add value.

C. Why massive transfusion results differ

On transfusion, the strongest model is the one that compresses most, reducing the entire window to nine values and classifying with a regularised linear model. The possible explanation for this is that when the recurrent classifier is replaced with logistic regression, the model is reduced from roughly 1700 fitted parameters to 10, estimated from around 75 positive training cases. Under the scarcity of training data, the variance from fitting a recurrent model might outweigh the additional structure that it could capture.

Additionally, massive transfusion may depend more on how the patient looks on average over the window (a persistently high heart rate or persistently low blood pressure). Averaging keeps that pattern and smooths the noise around it. Mortality may depend more on the out-of-the-ordinary moments (a heart rate suddenly dropping), which averaging removes.

D. Measurement presence as a confound

A model given only the binary mask of which channels were recorded reaches 0.718 on mortality and 0.662 on transfusion. The decision to attach an invasive blood pressure channel, for example, is a clinical judgement made about a patient the team already believes to be critically unwell. Therefore, any model trained on this data is at risk of recovering clinical assessment rather than the patient's state.

Concatenating the physiological channels together with the presence does not aid the models. It improves on physiology alone in two of six conditions and degrades mortality performance in all three. This is likely because concatenation increases the per-interval input dimension from 9 to 20, while the number of positive cases remains the same, so the classifier has more parameters to estimate from the same evidence.

E. Calibration and clinical usefulness

How useful the model can be practically depends not only on the AUROC but also on how rare the outcome is. Massive transfusion occurs in roughly 18% of the labelled cohort, so most patients do not need it. If the model is set to catch a high enough proportion of those who do, then most of the patients it flags will not have needed the blood. Whether that is acceptable is a clinical judgement about the relative cost. A missed activation leaves the ER unprepared for a patient who is rapidly losing blood, while a false activation mobilises staff who might be needed elsewhere. The models presented here are best understood as inputs to that judgement rather than as replacements for it.

F. Relation to prior work

The accuracies reported by Sigle et al. (2026) and Weidman et al. (2025) exceed anything achieved here, but both use injury mechanism, injury pattern and documented vital signs alongside the physiological stream on larger cohorts. The continuous record, therefore, appears to complement those variables and has a plausible role as an additional input to a model of the kind described by Pisirir (2024) or Marsden et al. (2025).

G. Legal, social, and ethical considerations

The data used here are retrospective clinical records from patients who were, by definition, unable to consent at the point of collection. The extracts analysed were anonymised before release: no direct identifiers are present, and no record can be traced to an individual patient. Timestamps were also shifted to the same nominal instant, thereby removing the incident's date and time from the data entirely.

The main issue that remains is one of interpretation. A model of this kind is intended to inform a clinician's judgement, not to replace it, but a numerical risk presented at the point of decision is easily read as a verdict rather than as one of the inputs. This matters most where the model is weakest: the mask-only results show that part of what is predicted reflects monitoring decisions already made by the crew, so an output treated as independent evidence would, in fact, partly echo the team's own assessment back to them. How such a prediction should be presented and what training would accompany it are left to future work.

VIII. Limitations and Future Work

The most important limitation of this study is the amount of labelled data. Mortality is evaluated on 84 test patients containing 15 deaths, and massive transfusion on 91 patients containing 18 positive cases. The confidence intervals around any single AUROC are therefore considerably wide, and no pairwise comparison between conditions can be tested for significance at this sample size. The results support ordering between conditions rather than the precise values of individual estimates, and differences of 2 or 3 points in AUROC should not be interpreted as meaningful. The same constraint applies to the LSTM tuning process, where the hidden width is chosen by averaging the validation AUROC across six seeds on 90 validation patients, each containing roughly a dozen positive cases.

A second limitation concerns how the conditions were compared. The whole-window convolutional pipeline differs from the others in both the encoder architecture and the downstream classifier, so its advantage in massive transfusion cannot be attributed to representation alone. On the same theme, the choice to evaluate lactate using the whole-window encoder was made after observing the model's performance on the test set; therefore, the analysis should be considered exploratory rather than pre-specified.

A third arises from the data pipeline itself. The analysis in Section III indicates that interior valid zeros are informative observations, yet zero readings were discarded throughout. This produced a cleaner analysis, but it may also have removed a useful signal.

Several of these constraints point directly to the work that should follow. The most important dimension left unaddressed is explainability, which any system intended to influence clinical decisions will require. Without attribution, a user cannot tell whether a prediction depends on the patient's physiology or on the fact that someone chose to attach an arterial line. Permutation importance at the channel level, attention weights over the twenty intervals, or SHAP values computed on the latent features would be an appropriate first step in that direction, and the results should then be evaluated with clinicians themselves, to assess both their usefulness and the adequacy of the explanations offered.

The comparison between encoders could also be strengthened. Only the latent dimension was swept here; depth, layer width, kernel size, and regularisation were fixed at plausible values, so what is compared are three specific configurations rather than three families of models. More extensive tuning would show whether the observed ordering in performance can be attributed to the architectures or to the chosen settings. One direction was started but not finished: a recurrent autoencoder in which an LSTM encoder compresses the window into a single hidden state and a decoder reconstructs the sequence from it, which would serve as an appropriate comparison to the convolutional architecture.

Finally, the Tiny Time Mixers models are worth testing. They are lightweight time-series models pre-trained on large external corpora that directly address the label scarcity that constrains every model in this study.

IX. Conclusion

This work asked whether the continuous physiological records collected during prehospital transport can support outcome prediction, and whether compressing those records into a compact latent summary makes them more usable. Discrimination was moderate for in-hospital mortality, weak for massive transfusion, and absent for arrival lactate, and compression was not beneficial. On mortality, the raw normalised channels outperformed all encoders, while on massive transfusion, the most aggressively compressed representation was the strongest model in the study.

A classifier given no physiological values at all, only the record of which channels were being measured, reached 0.718 on mortality and 0.662 on transfusion. On mortality, the raw channels and the dense encoder substantially outperform this baseline; the remaining models do not, and on transfusion only the whole-window encoder does. Any system built on data of this kind is therefore partly modelling the clinician's decision to monitor rather than the patient's condition.

Taken together, these findings position the continuous monitoring stream as a plausible additional input to existing trauma prediction models, contributing temporal detail that a single documented set of vital signs cannot. Establishing that would require the attribution methods this work did not implement, a broader search of other encoder architectures, and evaluation with clinicians themselves.

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XI. Appendix

A. Figures

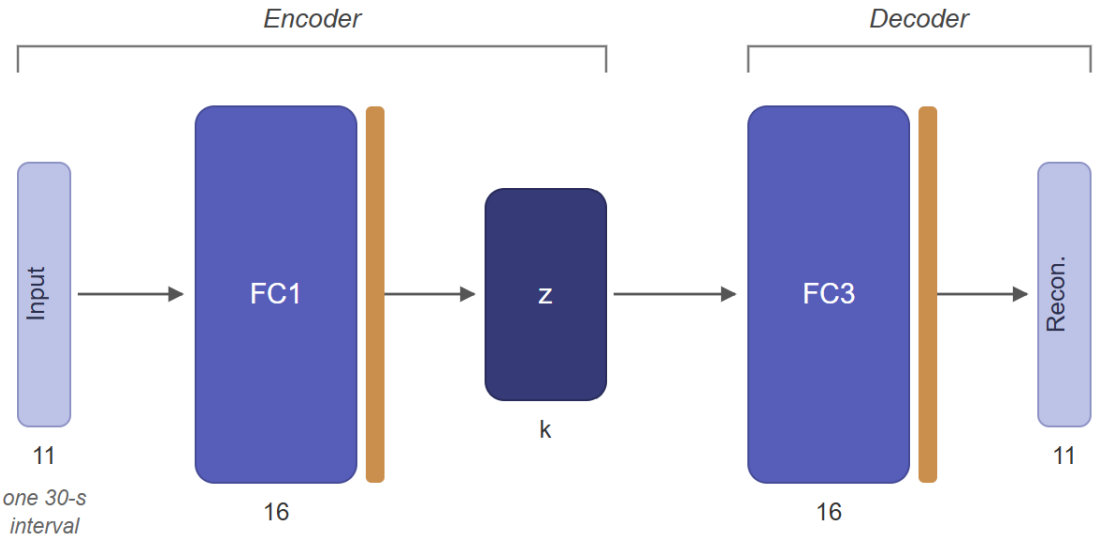


Fig 1. Architecture of a fully-connected encoder

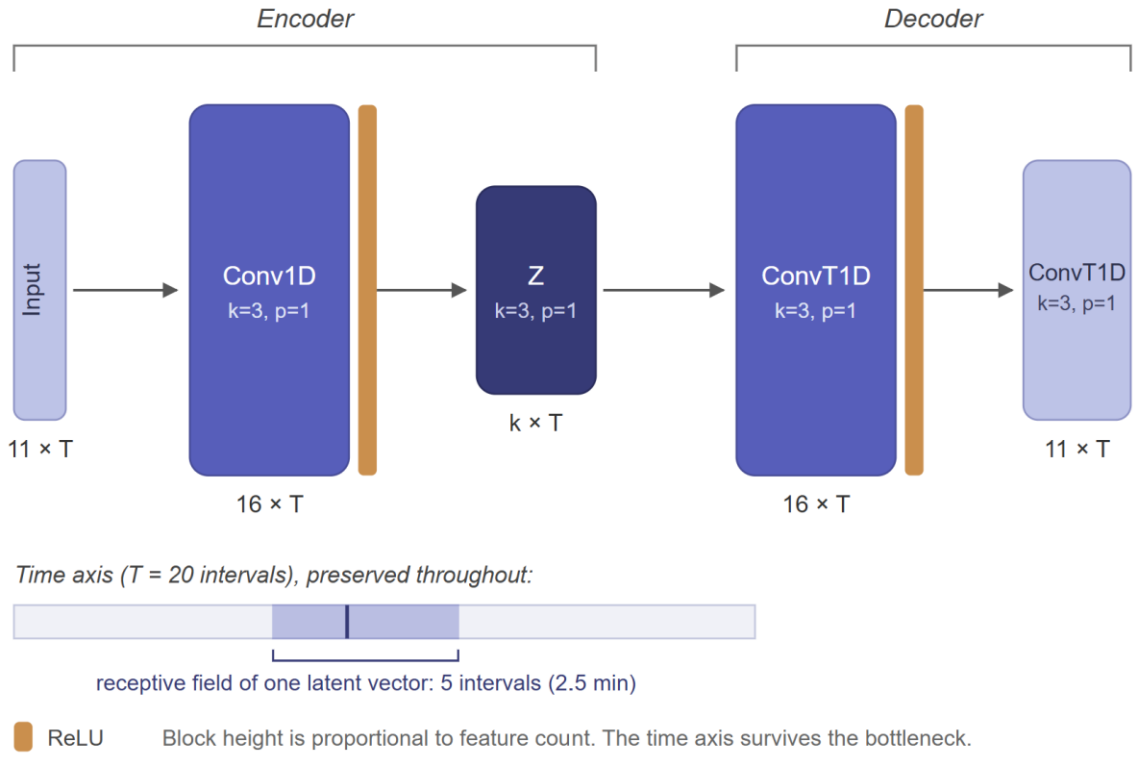


Fig 2. Architecture of a per-interval convolutional encoder

B. *Interior valid zeros analysis with head injury*

TABLE VI.
PROPORTION OF CAPNOGRAPHY RECORDINGS CONTAINING INTERIOR VALID ZEROS, BY HEAD-INJURY SEVERITY

Head AIS band	N	With zeros	%
0 - no head injury	224	65	29%
1-2 - minor/moderate	59	21	36%
3-4 - serious/severe	67	28	42%
5-6 - critical	65	29	45%

[c.] Head AIS was available for 416 of the 421 patients with capnography.

Patients with severe head injuries require the most intensive airway and ventilation management, and capnography is central to that management. The relationship with head-injury severity was therefore examined directly. The proportion rises steadily with severity, from 29% of patients with no head injury to 45% of those with critical head injury. This suggests the dropouts are not spread randomly across the cohort, but occur more often in patients receiving the most intensive airway management.

C. *Calibration*

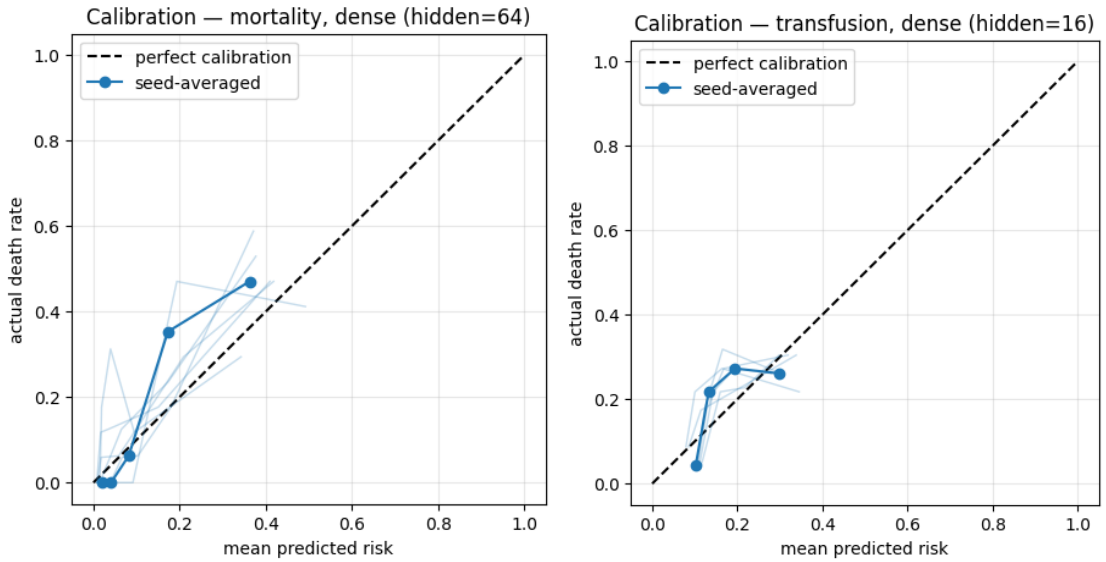


Fig 3. Calibration curves of a fully-connected autoencoder + LSTM

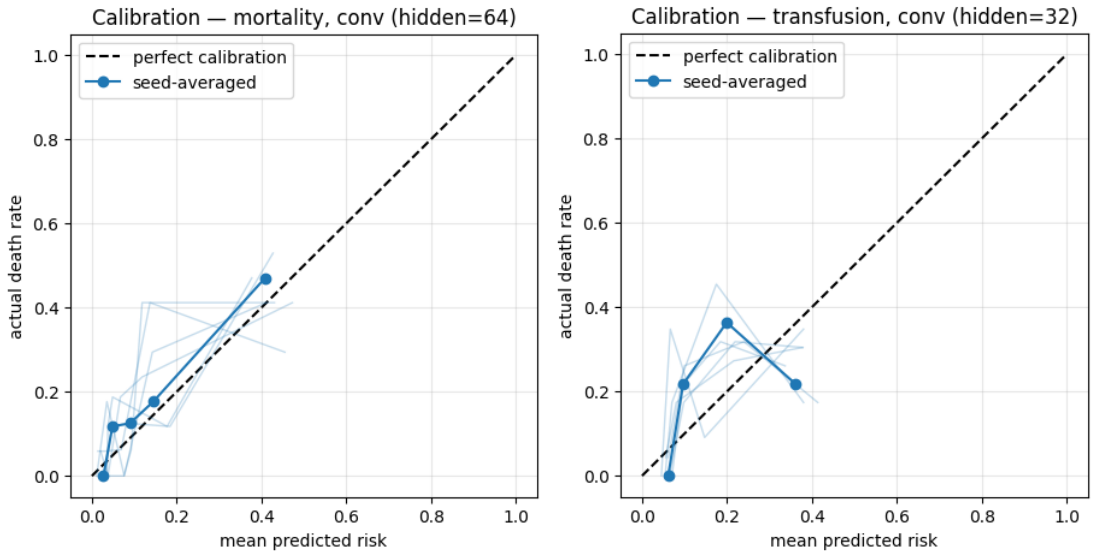


Fig 4. Calibration curves of a per-interval convolutional autoencoder + LSTM

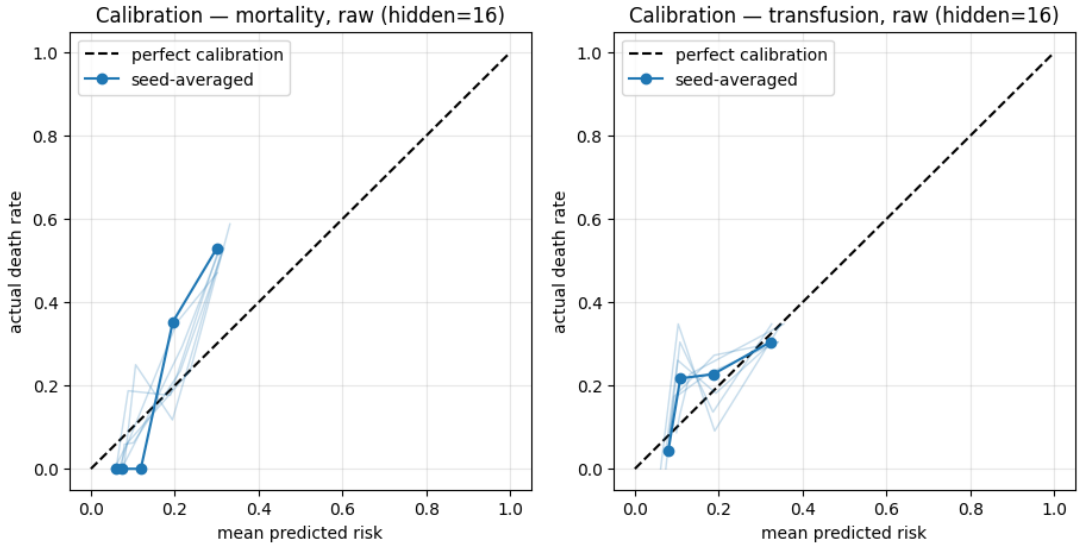


Fig 5. Calibration curves of raw channels + LSTM

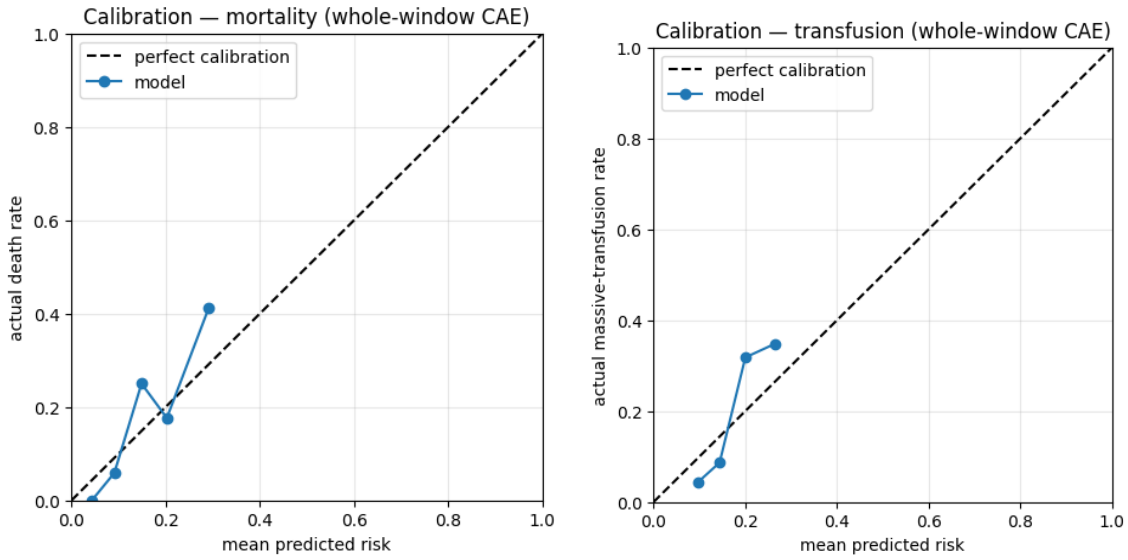


Fig 6. Calibration curves of a whole-window convolution autoencoder + Logistic regression

MSc Project - Reflective Essay

Project Title:	Deep Learning in Emergency Life-Saving Care: Continuous Prehospital Physiological Monitoring Data for Trauma Outcome Prediction
Student Name:	Marharyta Dzhafarova
Student Number:	[redacted]
Supervisor Name:	
Programme of Study:	MSc Data Science and Artificial Intelligence (Conversion)

This project asked whether the continuous physiological record produced by a ZOLL X monitor during prehospital transport can support prediction of trauma outcomes. The record is already collected for every patient and rarely used, largely because it is irregularly sampled and heavily incomplete. Three masked autoencoders and a raw-channel baseline were compared across three outcomes, with the presence pattern of the measurements held out as a separate experimental condition.

Strengths

The first strength is the architecture. Although the individual layers are standard, the objective and the bottleneck geometries are not. A masked autoencoder computes reconstruction error only where the mask indicates a genuine observation, normalised by the number of observed entries, so unmeasured positions contribute no gradient, and no imputation is required. I then designed three encoder–decoder pairs around that objective, deliberately choosing architectures that differ in the structure of their bottleneck: per-interval compression with no temporal context, per-interval compression over a five-interval receptive field, and compression of the entire window to a single patient vector. The whole-window decoder was the hardest of these to get working, since it expands a nine-dimensional vector back to a full twenty-by-eleven window, which is a substantially harder reconstruction problem than the per-interval ones.

The second is the experimental design around missingness. Che et al. (2018) established that missingness in clinical series carries information, and most subsequent work treats this as a feature to exploit. The design here treats the mask as a separate condition and asks whether the physiological values add anything beyond what the presence pattern already provides. This is not a widely tested framing; however, it produced some consequential results. The mask alone achieved test AUROCs of 0.718 for mortality and 0.662 for transfusion, compared with the best physiology figures of 0.857 and 0.730. Attaching an arterial line is a clinical judgement about a patient the crew already believes to be critically unwell, so a model trained on this data can recover the clinician's assessment rather than the patient's state. Without that condition, the four physiology models would have looked like straightforward successes; with it, only two clear the bar substantially for mortality, and only one does for transfusion.

The third is breadth. The study compares four pipelines that differ substantially from one another. Each was evaluated across the same three input conditions, six seeds, validation-selected hyperparameters and held-out test patients. This is what made it possible to see that the ordering reverses between tasks: raw channels are strongest for mortality, while the most aggressively compressed representation is strongest for transfusion. A study built around one architecture would have concluded that compression either does or does not help, when the honest answer is that it depends on which structure the outcome rests on.

Weaknesses

The clearest weakness is the depth of hyperparameter tuning. Three quantities were searched for: the latent dimension; the LSTM hidden width over {16, 32, 64}, selected based on validation AUROC; and the L2 penalty of the logistic regression, selected via cross-validation. Everything else was fixed early: the number of convolutional layers, the feature maps, the kernel width, the learning rate, the number of epochs, the batch size, and the absence of dropout or weight decay. The consequence is a limit on what the comparison produces: what was compared is four configurations, not four families of models, and a claim such as "convolutional encoding loses signal on mortality" should properly read "this convolutional encoder, at this depth and width, lost signal on mortality". A limited search does not simply make every model look slightly worse than it could be; it can change which model performs best. The whole-window

encoder is the most likely model to fall victim to this because it has the greatest architectural freedom to exploit.

Two further weaknesses should be recorded. The whole-window comparison is confounded because a single patient-level vector cannot be consumed by a recurrent model; thus, the pipeline changes the encoder and the classifier together, and its transfusion advantage cannot be attributed to the representation alone. A fairer design would have also run the sequence encoders through a pooled logistic head. Additionally, the labelled sample is small: 84 test patients with 15 deaths for mortality, and 91 with 18 positives for transfusion. No pairwise comparison can therefore be tested for significance, and differences of two or three AUROC points carry no weight. That was a constraint rather than a choice, and I have written the paper emphasising the ordering between conditions rather than the evaluation numbers.

What I would do differently

If I began again, the change would be to the order in which experiments were run. I would spend considerably less time on the fully connected encoder with an LSTM. It was built first because it was the simplest thing that could work, and it performs well at 0.810 on mortality, but it is the least interesting pipeline in the study. Much of the early effort went into it, and most of what that produced was infrastructure the later pipelines reused rather than findings.

More importantly, I would run the raw-channel baseline first. It is the cheapest experiment in the project, since there is no encoder to train, and it is the one that determines whether the rest of the study has a premise. Raw channels reached 0.857 on mortality, ahead of every encoder. Had that been known in the first weeks, the framing would have shifted from “which compression works best” to “under what conditions does compression help at all”, which is the question the results eventually forced anyway. The underlying error was one of attachment: I treated the autoencoder as the object of study rather than as a candidate solution to the stated problem, and so implemented the baseline last because it was conceptually the least interesting. The time recovered would have gone to the whole-window encoder, where further search had the best chance of changing the performance.

Further work

The most direct extension is a recurrent autoencoder. What distinguishes convolutional from recurrent encoding is the inductive bias. A convolution applies the same filter at every position and so detects local, translation-invariant structure over a bounded receptive field, whereas a recurrent encoder accumulates a state and is sensitive to order over an unbounded range (Bai, Kolter and Koltun, 2018). Moreover, in the whole-window pipeline, the sequence is destroyed not by the convolutions but by the adaptive average pooling that follows them. The correct counterpart is therefore an LSTM autoencoder compressing the window to its final hidden state (Srivastava, Mansimov and Salakhutdinov, 2015), an approach whose representations have been shown to transfer to downstream clinical prediction (Lyu et al., 2018).

The second extension is a proper hyperparameter search focused on the whole-window encoder, over depth, feature maps, kernel width, and latent dimension. The third is pre-trained time-series models, with Tiny Time Mixers (Ekambaram et al., 2024) as the most promising candidate. Label scarcity is the binding constraint for this study and cannot be addressed by architecture in a cohort of this size. A model pre-trained on large external corpora imports structure that 606 patients cannot supply, and remains small enough to be practical.

The fourth is explainability. Nothing in the current pipelines indicates to a user whether a prediction relies on the patient’s physiology or on the crew’s decision to attach a channel, and the mask result shows that both contribute. That gives explainability a well-defined role here: attribution computed separately over the physiological channels and over the mask would tell a clinician how much of the risk estimate is independent evidence and how much is their own team’s judgement. The difficulty here is not the classifier but the representation: the whole-window pipeline ends in a logistic regression whose coefficients are perfectly interpretable over nine latent dimensions that mean nothing to anyone. This is also a regulatory requirement rather than only a design preference. Under Regulation (EU) 2024/1689, AI intended as a safety component of a medical device is high-risk and carries obligations on transparency, human monitoring, logging and risk management, with comparable expectations set out in the UK by the MHRA (2022). A system that cannot report what a prediction was based on cannot support the oversight those frameworks require.

Finally, explainability should be assessed with the users in mind. This can present prehospital and emergency clinicians with candidate output formats – a risk figure alone, a risk figure with per-channel attribution, and a display separating physiological from monitoring-pattern contributions. And later, asking through an interview and a structured survey which they can act on under time pressure and which they would trust. Beede et al. (2020) showed how wide the gap can be between a model that performs well in evaluation and one that functions in a clinic.

The relationship connecting theory and the practical work

The theoretical case for the pipeline held only partly. Autoencoders are motivated on the grounds that a bottleneck discards variance that does not aid reconstruction, retaining structure and shedding noise (Hinton and Salakhutdinov, 2006), and Zhao et al. (2019) proposed pairing a convolutional autoencoder with a downstream LSTM specifically because reconstruction smooths high-frequency content so the recurrent model sees cleaner input. What failed to transfer was not the theory of autoencoders but an unexamined assumption inside it: that high-frequency content is noise. In traffic-flow and industrial-sensor data, it most probably is, whereas in a ten-minute prehospital window, an abrupt change in heart rate is not a nuisance fluctuation but may be the event the model is meant to detect.

Theory did deliver on another part of the project. The three-condition ablation follows directly from Che et al. (2018): if missingness is informative, a physiology model evaluated without a missingness control cannot be interpreted because there is no way to tell how much of its performance is attributable to physiology. That decision was made before any results were produced.

Legal, social, ethical and sustainability considerations

The data are retrospective clinical records from patients who, by the nature of their injuries, were unable to consent at the point of collection. The extracts released for analysis were anonymised beforehand, with no direct identifiers present and no record traceable to an individual, and timestamps were shifted to a common nominal instant, removing the date and time of each incident entirely. Under the UK GDPR and the Data Protection Act 2018, effectively anonymised data falls outside the definition of personal data, and the project accordingly did not require ethical approval.

The ethical question that most concerns me is interpretation at the point of use. A model of this kind is intended to inform a clinician's judgement rather than replace it, but a numerical risk presented during a time-critical decision is easily read as a verdict rather than as one input among several. The mask result sharpens this considerably. A model that partially recovers the crew's own monitoring decisions returns their assessment to them. The failure is also distributed in the worst direction: the mask signal is strongest where monitoring was most intensive, so the model is best informed about patients the crew had already identified as critically unwell, and least informed about the patient whose deterioration went unrecognised – who is exactly the patient a decision support tool is meant to catch.

On sustainability, the models developed here are small, and I regard this as a genuine strength. The encoders comprise a few thousand parameters each and the largest downstream LSTM is on the order of twenty thousand, so the complete study trains within minutes to hours on a single ordinary machine, if no pre-training on external corpora is involved.

Reflection on the project process

The lesson I take from this project is about the process rather than technique. The work I am most pleased with (the masked reconstruction objective, the three-condition ablation, the interior-zeros analysis) is work in which I asked what the data actually were before deciding what to do with them. The weakness was committing to an approach early and then testing it, rather than establishing the cheap constraints first and letting them determine where the expensive effort goes. That is a transferable lesson, and one I expect to carry further in my research work.

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ECS750P/ECS753P/ECS754P– MSc Project Module
Generative AI Tools – Student Accountability Statement

The completed and signed statement must be included in the appendix of the final dissertation paper.

Student Name: Marharyta Dzhafarova
Project Title: Deep Learning in Emergency Life-Saving Care: Continuous Prehospital Physiological Monitoring Data for Trauma Outcome Prediction
Supervisor Name: William Marsh

1. Have you used Generative AI tools for your project?

YES
NO

If yes, please complete sections 2 and 3 and sign the declaration below.
If no, please sign the declaration below.

2. Please provide details of all the AI tools that you used in your project work.

AI Tool	Version	Publisher	Access URL
Claude	Opus 4 and 5	Anthropic	https://claude.ai/
Grammar checker		Grammarly	https://grammarly.com/

3. If you have used Artificial Intelligence (AI) tools in any part of your project please include in table format how, where, and why these tools were used. This should include references to the relevant parts of your project work — for example:

- Specific chapters or sections of the report (e.g. literature review, methodology, requirements)
- Software development (e.g. AI-assisted code generation, debugging, documentation)

- Interview or survey materials (e.g. AI-generated or refined questions/prompts)
- Datasets (e.g. AI-assisted data cleaning, labelling, or synthesis)
- The showcase video (e.g. AI-assisted scripting, narration, or editing support)

Be specific and transparent about your use of AI tools, including both direct outputs (e.g. text, code, images) and indirect uses (e.g. idea generation, editing assistance). The goal is to ensure academic integrity and provide a clear record of how AI contributed to your work.

Project Section	Description	Justification
Literature Review	I used it to provide an initial list of papers that would be relevant for my topic.	I used GenAI to speed up the search process but I looked into every article mentioned in the study myself.
Methodology	I used it to explain to me the concepts of the fully-connected autoencoder, convolutional autoencoder and a LSTM, including generating visualisation that significantly improved my understanding of the models. I used GenAI to debug my code and structure it into functions.	I used GenAI for explanation but the choices of the architectures I made myself consulting with my supervisor. Because of the complexity of my experiment, I needed my code to be reproducible with different architectures. The final judgement on the implementation of certain functions or short cuts was always made by me.
Dataset	I used GenAI to debug my code and structure it into functions in the exploratory data analysis part.	Because my code needed to work on 11 different measures I needed to make it as universal as possible and that is where I used GenAI to improve data cleaning and windowing functions.
The write-up	I used Grammarly and GenAI to improve the structure of certain sentences as well as improve the flow of some sections. I used Grammarly to check for plagiarism as well.	I used it as a writing support tool, not a substitute. Every final decision regarding the thesis structure and content was made by me.
Video presentation	I used GenAI to assist generating visualisations in PowerPoint from my written descriptions.	Because of the technical complexity of my pipelines, visualising the architectures was necessary for the second grader's understanding of my work. GenAI has significantly sped up the process of generating those images.
Evaluation	I used GenAI to evaluate my thesis against the grading scheme.	To make sure that I didn't miss any important points when it comes to text and referencing structure.

3. Generative AI Student Declaration

- I have declared all uses of AI tools (e.g., coding assistants, text generators, data analysis tools) in this project.
- I have reviewed and verified all AI-assisted content to ensure accuracy and suitability.
- I take full responsibility for the originality and integrity of my project submission.
- I have referenced all sources, including those suggested or produced with AI support.
- I have not used AI in ways that undermine academic honesty (e.g., plagiarism, unverified code submission).
- My use of AI was supplementary and did not replace my own independent learning and problem-solving.

Signed:  Date: 18.08.2026